Value investing: buy-and-hold results

S&P 500, data as of 2026-10-31

Abstract

Each January and July since 2016, a rule-based value score picks 5 S&P 500 companies, and 100 USD is invested in them, weighted by score and never sold. Investors who started in 2016, 2020, 2023 and 2025 are followed to 30 October 2026, each against the same money in the S&P 500 total-return index and in Berkshire Hathaway. The reference is the investor who started in 2020: the data hold only today's index members, and 125 companies have left the index since then against 230 since 2016, so fewer failed companies are missing. That investor's 1,400 USD is worth 4,700 USD, against 2,480 USD in the S&P 500 and 2,415 USD in Berkshire Hathaway; a second wallet that also sells holdings whose score has fallen is worth 4,681 USD. The report closes with the next picks: First Solar (FSLR), Cognizant (CTSH), Snap-on (SNA), Skyworks Solutions (SWKS) and ResMed (RMD). The result rests on few decisions and few companies, with 50% of the value in one company; it is a research result, not evidence of a lasting edge.

1 Method

Company fundamentals are the values first reported in each annual report (10-K) to the SEC, used from the day they were filed. When a company reports an item under more than one label, the label it used most recently by the date is taken, so later filings never decide it (Appendix D). Prices and dividends are from Yahoo Finance; returns include reinvested dividends and are in US dollars. On each date, only companies already in the S&P 500 are considered, outside the Financials and Real Estate sectors.

Sixteen rules from Graham, Buffett and a net income growth test grade every company. For each rule, the companies are ranked by the rule's value and placed in six equal groups, from 5 points for the best sixth to 0 for the worst; each rule's points are multiplied by its weight, one less the share of companies that pass the rule's threshold, so rules few companies pass count more. The score is the sum. On the first trading day of January and of July, the 5 highest-scoring companies are bought for 100 USD, each in proportion to its score.

Three limits apply to every number in this report. Companies that left the index are missing from the data, which flatters the results. The score's design and every setting of this report (the rules' weighting, the five picks, the decision months, the sell threshold, the start years) were chosen while looking at the whole period from 2016, so no period here is out of sample: the results describe how these rules would have done, not evidence that they will do so. There are no trading costs or taxes.

2 Picks per half-year

Table 1: Purchases per half-year and their value today (USD).

Bought	Picks and weights	Wallet	S&P 500	Berkshire
2016-01	WDC 21%, CMI 21%, GLW 21%, CSCO 20%, NTAP 17%	821	452	415
2016-07	WDC 22%, CSCO 21%, NTAP 19%, QCOM 19%, CMI 19%	756	428	377
2017-01	CSCO 22%, QCOM 21%, LRCX 20%, CTSH 19%, CMI 18%	781	394	331
2017-07	GLW 21%, QCOM 21%, CSCO 20%, CTSH 20%, BAX 19%	305	363	316
2018-01	CSCO 21%, TPR 20%, GLW 20%, CTSH 19%, QCOM 19%	315	324	275
2018-07	QCOM 22%, TPR 21%, CSCO 19%, CTSH 19%, SNA 19%	273	317	289
2019-01	MU 22%, LRCX 21%, CTSH 21%, KLAC 18%, SNA 18%	1,441	341	267
2019-07	MU 21%, NUE 21%, SNA 20%, CTSH 20%, LRCX 18%	925	286	253
2020-01	NUE 21%, CTSH 21%, MU 20%, SNA 20%, CMI 18%	579	257	238
2020-07	CTSH 21%, NUE 21%, HAS 20%, MU 19%, SNA 18%	638	266	305
2021-01	HAS 21%, NUE 21%, CTSH 20%, MU 20%, HUM 19%	429	223	237
2021-07	SNA 21%, INTC 20%, CMI 20%, MU 20%, HUM 19%	351	189	194
2022-01	MU 21%, SNA 20%, CMI 20%, HUM 20%, INTC 19%	329	169	180
2022-07	NUE 22%, SNA 20%, MU 20%, INTC 19%, CMI 19%	526	211	195
2023-01	MU 22%, NUE 21%, CTSH 19%, INTC 19%, SNA 19%	535	209	175
2023-07	STLD 22%, NUE 21%, MU 21%, SNA 19%, PKG 18%	439	178	159
2024-01	STLD 22%, NUE 21%, SWKS 20%, SNA 19%, CTSH 18%	141	166	150
2024-07	NUE 22%, STLD 20%, SWKS 20%, SNA 20%, BG 19%	144	142	134
2025-01	NUE 21%, SWKS 21%, STLD 20%, BG 19%, AMAT 19%	191	132	120
2025-07	SWKS 22%, ON 21%, SNA 20%, AMAT 19%, STLD 19%	154	124	111
2026-01	SWKS 21%, ON 21%, SNA 20%, NUE 19%, CTSH 19%	123	112	109
2026-07	CTSH 21%, SWKS 20%, SNA 20%, RMD 19%, BF-B 19%	122	102	109

Table 1 lists the companies bought each January and July, with their weights, and what each 100 USD is worth on 30 October 2026, next to the same money in the S&P 500 and in Berkshire Hathaway. Figure 1 shows the same values. Most of the money went to Information Technology, Industrials and Materials; the best single purchase is MU in January 2019, now worth 28 times its cost. The purchases of July 2017, January 2018, July 2018 and January 2024 are worth less today than the same money in the S&P 500. Appendix A lists every purchase of the wallet with sells of the investor who started in 2020, with its price at purchase and today, what later sales took from it, what it keeps today and its total return with dividends.

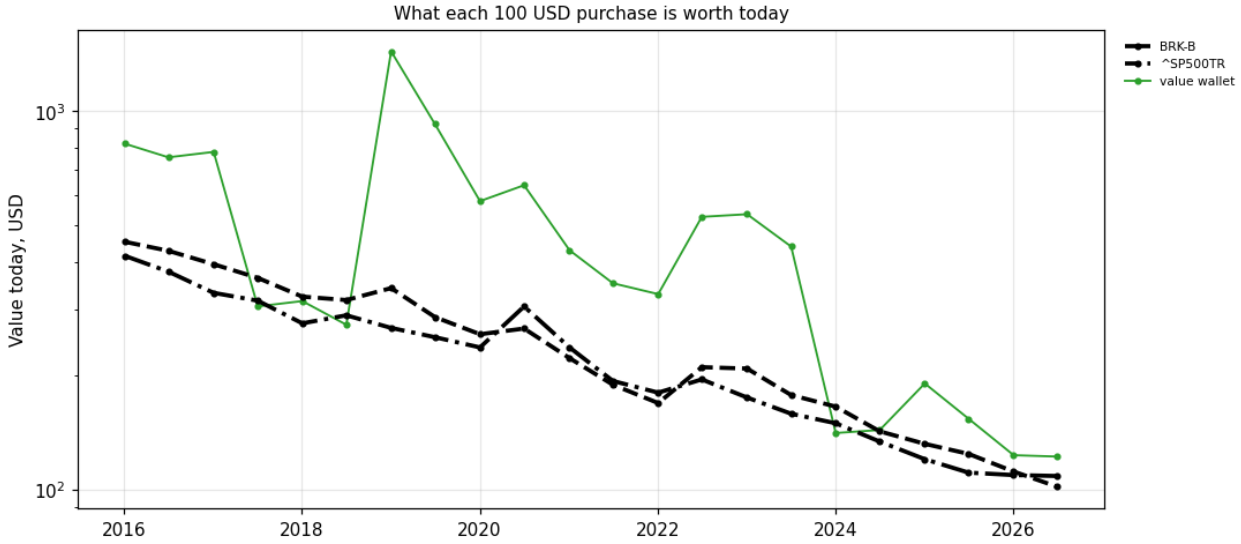


Figure 1: What each purchase is worth today, by date of purchase, on a logarithmic scale.

3 Value over time

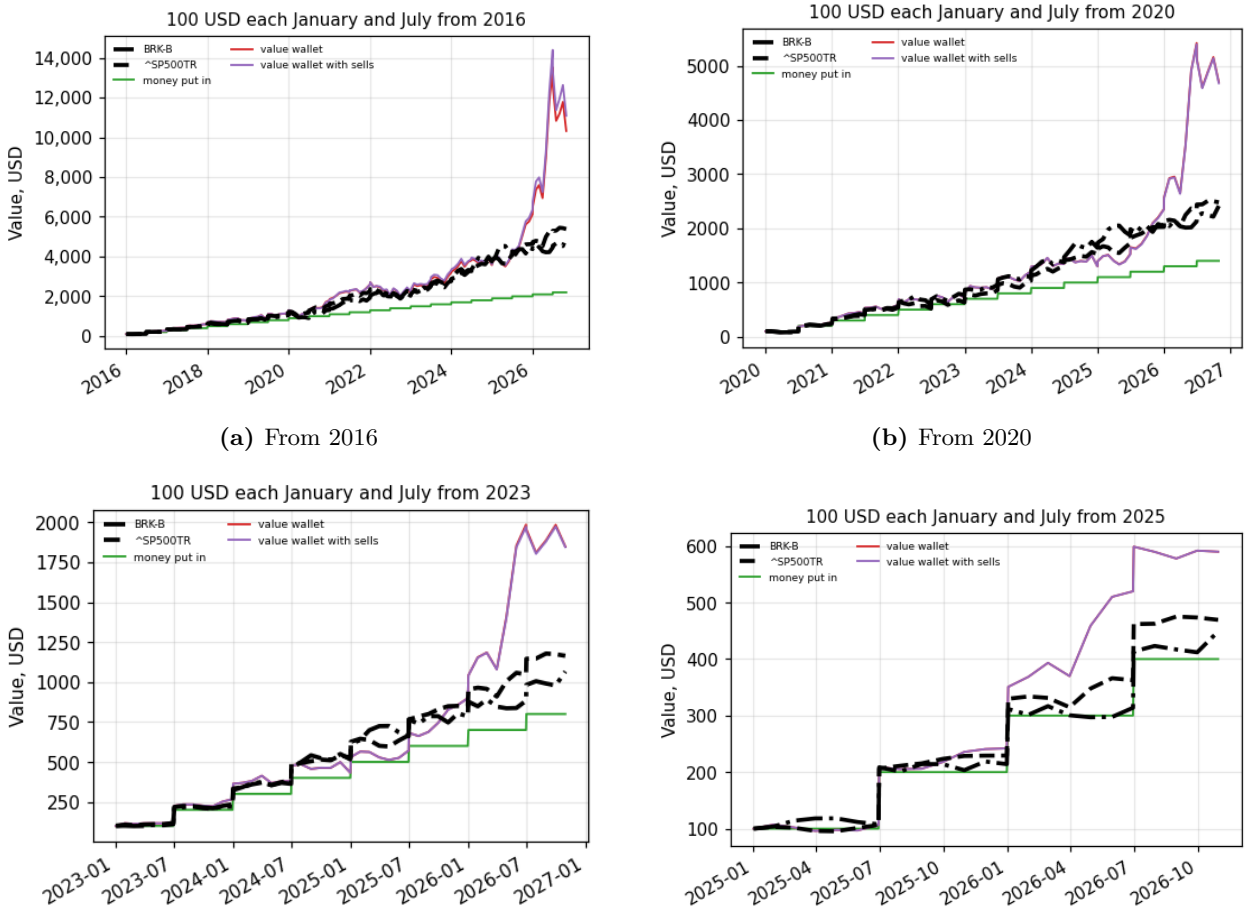
Table 2: Money put in and value today per start year (USD), and the money-weighted return a year.

Start	Holder	Put in	Value	Multiple	A year
2016	Value wallet	2,200	10,317	4.69	26.0%
	Value wallet with sells	2,200	11,086	5.04	27.2%
	S&P 500	2,200	5,383	2.45	15.3%
	Berkshire Hathaway	2,200	4,938	2.24	13.9%
2020	Value wallet	1,400	4,700	3.36	33.9%
	Value wallet with sells	1,400	4,681	3.34	33.8%
	S&P 500	1,400	2,480	1.77	15.9%
	Berkshire Hathaway	1,400	2,415	1.73	15.2%
2023	Value wallet	800	1,848	2.31	43.7%
	Value wallet with sells	800	1,845	2.31	43.6%
	S&P 500	800	1,164	1.46	18.7%
	Berkshire Hathaway	800	1,066	1.33	14.2%
2025	Value wallet	400	590	1.47	41.0%
	Value wallet with sells	400	590	1.47	41.0%
	S&P 500	400	469	1.17	15.7%
	Berkshire Hathaway	400	449	1.12	11.1%

Table 2 and Figure 2 follow the four investors. Each puts in 100 USD every January and July from the start year; the value wallet never sells. The return a year is money-weighted: the constant yearly rate that turns the payments into today's value.

The investor who started in 2016 earns 26.0% a year against 15.3% in the S&P 500 and 13.9% in Berkshire Hathaway. From 2020 the rates are 33.9%, 15.9% and 15.2%; from 2023 43.7%, 18.7% and 14.2%; from 2025, over less than two years, 41.0%, 15.7% and 11.1%. The wallet's value swings more than either comparison, because five companies of a few industries are far less diversified than the index or Berkshire.

The value wallet with sells buys the same picks with the same money, but at each decision it also sells part of any holding whose score has fallen; Section 4 describes the rule. From 2016 it is worth 11,086 USD (27.2% a year), from 2020 4,681 USD (33.8%) and from 2025 590 USD (41.0%).



(a) From 2016

(b) From 2020

(c) From 2023

(d) From 2025

Figure 2: Value of 100 USD every January and July from each start year, in USD, next to the money put in; the S&P 500 dashed and Berkshire Hathaway dash-dot.

4 Selling below the threshold

Table 3: Sales of the investor who started in 2020: share of each holding sold, its score on the 0-to-1 scale, and the proceeds.

Year	Sold: share of the holding (score)	Proceeds, USD
2023-07	HAS 11% (score 0.36)	3.85
2024-01	HAS 2% (score 0.39)	0.40
2024-07	HAS 60% (score 0.16)	17.72
2025-01	HAS 60% (score 0.16)	7.03
2025-07	HAS 15% (score 0.34), INTC 9% (score 0.36)	5.36
2026-01	HAS 14% (score 0.34), INTC 11% (score 0.36)	9.01
2026-07	HAS 40% (score 0.24)	2.05

Each January and July, before buying, the wallet with sells checks every company it holds. The score is put on a scale from 0 to 1, the share of the most the company could score on the rules it is measured on. A holding scoring 0.4 or less is sold in part: the share sold is (0.4 less the score) divided by 0.4, so all of it at a score of 0 and almost none just under 0.4. Holdings are sold, never sold short. The proceeds are added to the decision's 100 USD and buy its picks. Table 3 lists the 9 sales of the investor who started in 2020, worth 45 USD in all.

5 Wallet contents

Figure 3 follows every company in the wallet with sells of the investor who started in 2020, month by month. A line starts or steps up when the company is bought, steps down when part of it is sold, and rises or falls with the price in between. The dashed lines follow the same money in the S&P 500 and in Berkshire Hathaway, divided by the number of companies the wallet holds on the date, so they show what one holding would be worth in the index or in Berkshire. Appendix B gives each holding's value after every January's trades, the money put in and the proceeds of sales.

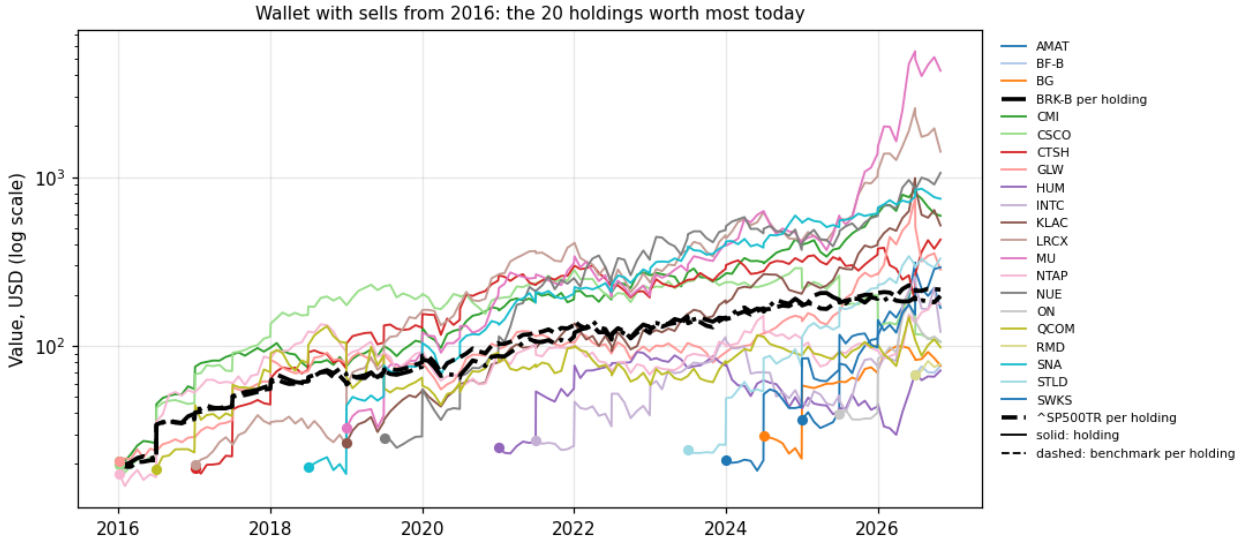


Figure 3: Value of each holding in the wallet with sells (solid) and of the S&P 500 and Berkshire Hathaway divided by the number of holdings (dashed), in USD on a logarithmic scale.

6 Rules behind the winners

Table 4: Dollars ahead of the same money in the S&P 500, per pick: total, average and median of the picks that beat the index and of those that lagged it.

Picks	Number	USD ahead, total	Average	Median
Beat the S&P 500	63	+5,601	+88.9	+17.4
Lagged the S&P 500	47	-667	-14.2	-8.1
All picks	110	+4,934	+44.9	+4.5

Of the picks, 63 have done better than the S&P 500 since their purchase and 47 have not, but a count treats a large win and a small loss alike. Table 4 weighs them in dollars: each pick's value today less what the same money would be worth in the S&P 500. The winners are 5,601 USD ahead of the index and the laggards 667 USD behind, a net 4,934 USD, the wallet's whole lead over the index. The median winner is 17 USD ahead and the median laggard 8 USD behind; the average win is 6.3 times the average loss. The wins are concentrated: the 5 largest, MU Jan 2019, LRCX Jan 2017, MU Jul 2019, MU Jan 2023 and LRCX Jan 2019, make up 37% of the winners' total.

Figure 4 compares the mean points per rule of the winners and of the laggards on the day they were picked, each pick weighted by its dollars ahead or behind the index, so the large wins and losses count most; the rules are ordered by how much more the winners scored. The winners scored most above the others on price to tangible assets (E6), P/E (G6), P/E times P/B (G7) and 4-year earnings growth (E5); the picks that lagged scored higher on 10 years of profit (G3) and return on equity (B1). With so few picks, these differences describe this sample; they do not establish which

rules work. Appendix E explains each rule, and Appendix C tests whether the rule weights favour the rules that told winners from losers.

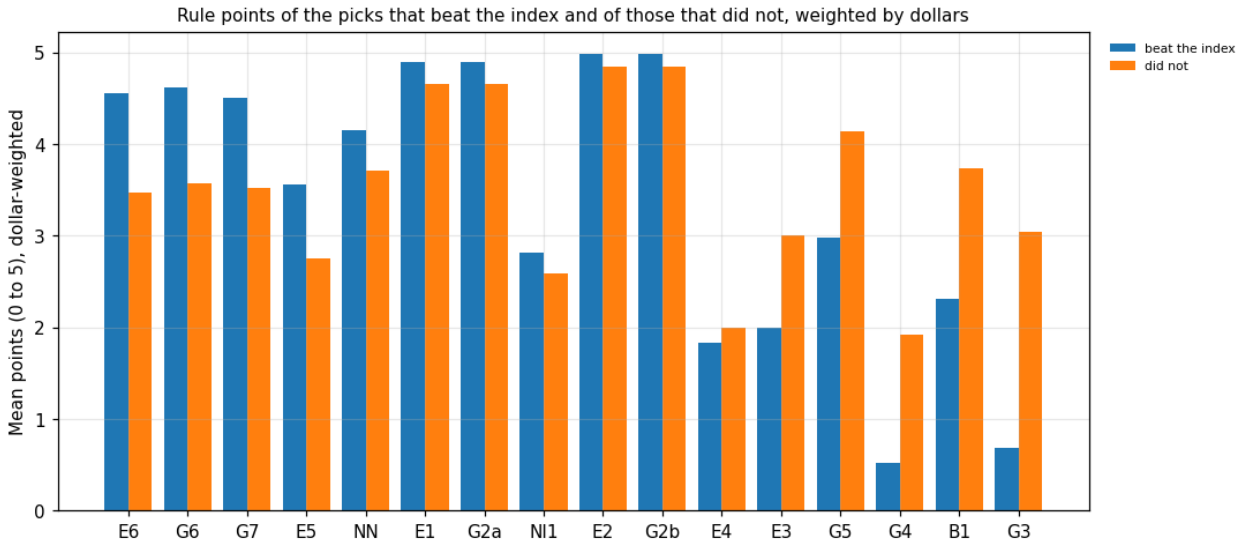


Figure 4: Mean points per rule (0 to 5) of the picks that beat the S&P 500 since purchase and of those that did not, each pick weighted by its dollars ahead or behind the index.

7 The Magnificent 7

Table 5: The Magnificent 7 on the score: score and score on the 0-to-1 scale on the latest score date, rank among the companies scored then, and median rank on the dates of the decisions since 2016.

Company	Score	Score 0 to 1	Rank today	Median rank
Nvidia (NVDA)	29.0	0.59	22 of 387	40
Microsoft (MSFT)	27.5	0.56	37 of 387	36
Meta Platforms (META)	25.2	0.77	52 of 387	192
Alphabet Inc. (Class A) (GOOGL)	24.2	0.56	68 of 387	28
Tesla, Inc. (TSLA)	18.6	0.40	153 of 387	149
Amazon (AMZN)	16.3	0.33	209 of 387	239
Apple Inc. (AAPL)	12.4	0.28	300 of 387	143

The best-known growth companies of the period, the Magnificent 7, are rarely picked. On the latest score date Nvidia (NVDA) ranks highest of them, 22 of 387 companies, and Apple Inc. (AAPL) lowest at 300 (Table 5). The rules are Graham's and Buffett's, written to avoid paying a high price for growth: Figure 5 shows that the Magnificent 7 score far fewer points than the picks on P/E times P/B (G7), P/E (G6), price to tangible assets (E6) and current dividend (E4), and more only on 4-year earnings growth (E5), net income growth (NI1) and 10-year earnings growth (G5).

The same 100 USD at each decision, split equally over the Magnificent 7 in the index on the date and never sold, is worth 22,613 USD from 2016 (39.0% a year), 4,445 USD from 2020 (32.3%) and 462 USD from 2025 (14.1%), against 10,317, 4,700 and 590 USD in the value wallet. The comparison picks the Magnificent 7 with hindsight: they are named for having been the winners of these years.

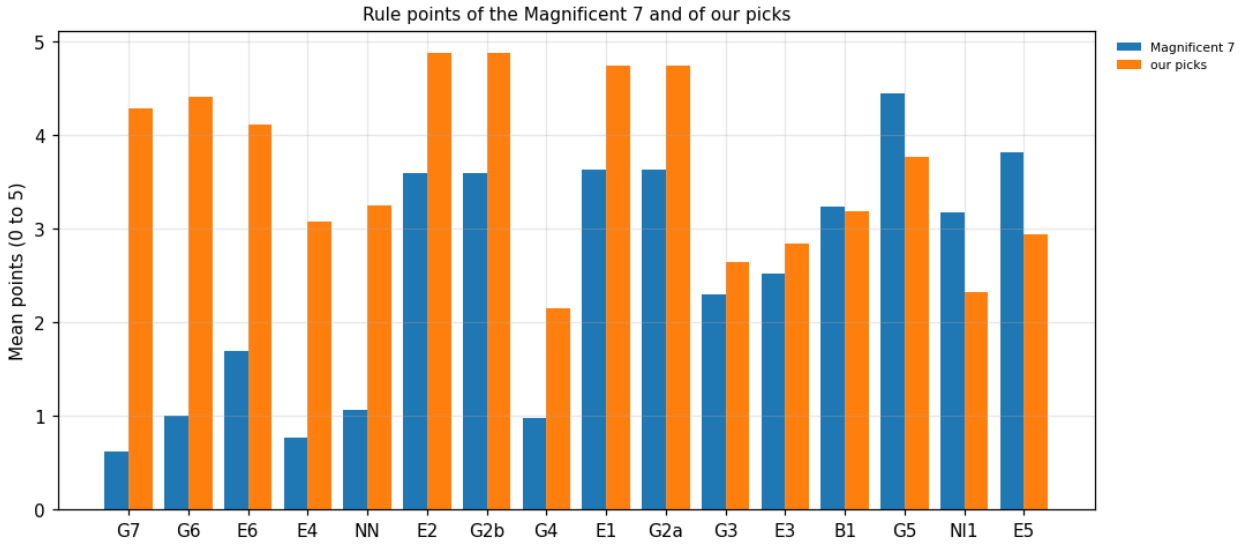


Figure 5: Mean points per rule (0 to 5) of the Magnificent 7 and of our picks on the decision dates since 2016, the rules ordered by how many more points the picks scored.

8 Price history of the picks

Figure 6 shows the share price of every company the investor who started in 2023 bought or sold, and of the next picks, over the last five years. The price is split-adjusted and excludes dividends, as a price chart shows it; the companies run from the best to the worst price change over the five years, given beside each name; when they do not fit on one page, only the five best and the five worst are shown. Each purchase is marked in green and each sale in red, at the price on the decision date; a larger marker means more money. Figure 7 gives the same trades in USD per company and decision date.

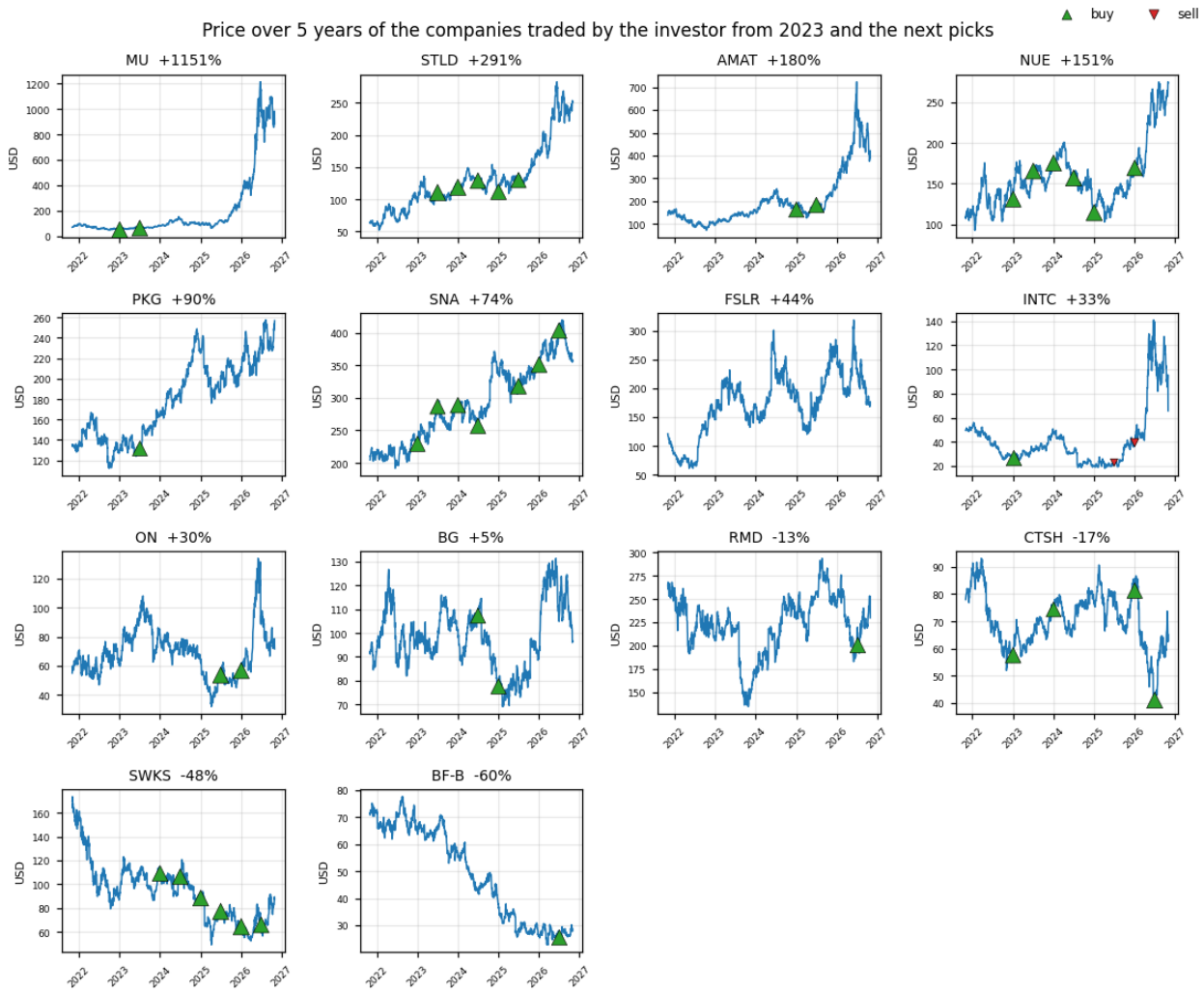


Figure 6: Share price (USD, split-adjusted, dividends excluded) over the last five years, one panel per company, ordered from the best to the worst price change over the period; purchases (green) and sales (red) of the investor who started in 2023, marker area in proportion to the USD traded.

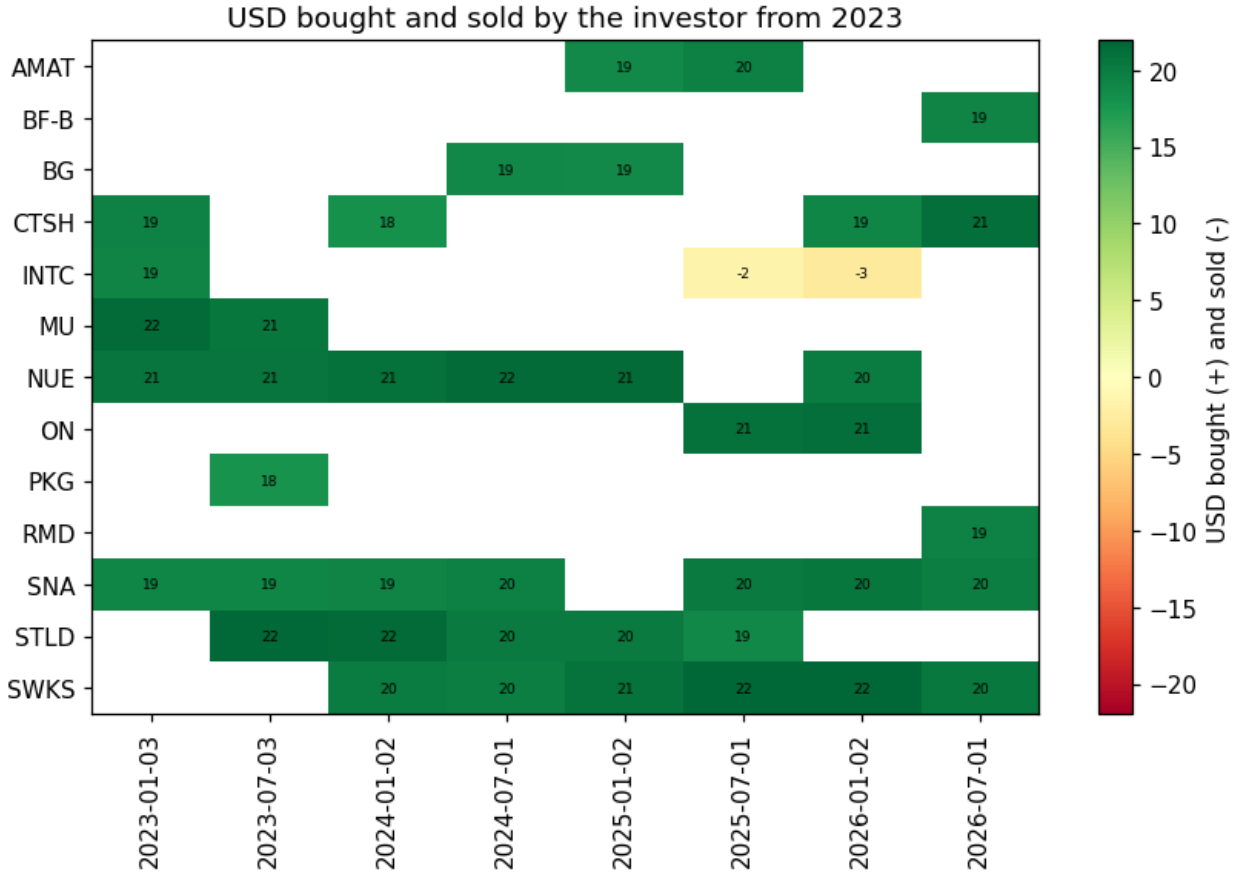


Figure 7: USD bought (green, positive) and sold (red, negative) by the investor who started in 2023, per company and decision date.

9 The next picks

Table 6: The next picks on the latest score date: score, score on the 0-to-1 scale, weight and USD of each 100 USD.

Company	Sector	Score	Score 0 to 1	Weight	USD
First Solar (FSLR)	Technology	39.2	0.83	20.5%	20.49
Cognizant (CTSH)	Technology	38.7	0.79	20.3%	20.26
Snap-on (SNA)	Industrials	38.1	0.85	19.9%	19.95
Skyworks Solutions (SWKS)	Technology	37.9	0.77	19.8%	19.80
ResMed (RMD)	Health care	37.3	0.76	19.5%	19.50

On the latest score date, 18 September 2026, the five highest-scoring companies are First Solar (FSLR), Cognizant (CTSH), Snap-on (SNA), Skyworks Solutions (SWKS) and ResMed (RMD) (Table 6). Each would receive its share of the next decision's 100 USD and of any proceeds from sales. The decision is taken on the first trading day of January with the scores known then, which include the annual reports filed until that day; these picks are a preview and may change. Figure 8 shows each pick's share price over the last five years, with the earlier purchases of the investor who started in 2023 and the potential purchase at today's price.



Figure 8: Share price of each next pick (USD, split-adjusted, dividends excluded) over the last five years: earlier purchases of the investor who started in 2023 in green, the potential purchase at the latest price in orange.

9.1 Sell check

Table 7: Holdings the rule would sell in part: value today, score on the 0-to-1 scale, share to sell and amount.

Company	Holding, USD	Score 0 to 1	Sell	Sell, USD
Intel (INTC)	111.96	0.39	4%	3.96
Hasbro (HAS)	4.53	0.24	40%	1.81

The investor who started in 2020 holds 16 companies. With the scores of 18 September 2026, the rule would sell part of Intel (INTC), 4% and Hasbro (HAS), 40%, and keep the other 14 as they are (Table 7).

10 Conclusions

In this sample, the value picks bought twice a year and held are worth more today than the same money in the S&P 500 or in Berkshire Hathaway for each of the four start years. That describes the past under rules and settings chosen while looking at it; it is not a test of the strategy. The lead is uneven: the 8 purchases of Micron Technology (MU) make up 50% of the 2020 investor's value, and the purchases of July 2017, January 2018, July 2018 and January 2024 trail the index.

Before the strategy is used, three steps are needed: the companies that left the index must be added to the data, the score must be tested on more decisions (each quarter instead of each year), and trading costs and taxes must be included.

A Every purchase

Table 8: Every purchase of the wallet with sells of the investor who started in 2020: weight in the decision's purchase, USD invested (the decision's 100 USD and the proceeds of sales), split-adjusted price at purchase and today, USD sold from it later, USD it keeps today and its total return with dividends, (sold + kept) / invested - 1.

Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2020-01	NUE	Materials	21%	21	55.12	273.87		120	+463%
2020-01	CTSH	Technology	21%	21	61.92	65.11		25	+17%
2020-01	MU	Technology	20%	20	55.39	884.74		326	+1538%
2020-01	SNA	Industrials	20%	20	169.00	355.66		50	+151%
2020-01	CMI	Industrials	18%	18	179.32	500.09		59	+227%
2020-07	CTSH	Technology	21%	21	56.42	65.11		27	+28%
2020-07	NUE	Materials	21%	21	40.38	273.87		156	+652%
2020-07	HAS	Discretionary	20%	20	74.69	119.09	18	2	+1%
2020-07	MU	Technology	19%	19	49.70	884.74		356	+1726%
2020-07	SNA	Industrials	18%	18	134.12	355.66		58	+212%
2021-01	HAS	Discretionary	21%	21	91.96	119.09	15	2	-20%
2021-01	NUE	Materials	21%	21	52.51	273.87		117	+468%
2021-01	CTSH	Technology	20%	20	79.42	65.11		18	-10%
2021-01	MU	Technology	20%	20	74.05	884.74		240	+1125%
2021-01	HUM	Health care	19%	19	412.19	414.50		20	+6%
2021-07	SNA	Industrials	21%	21	223.42	355.66		38	+83%
2021-07	INTC	Technology	20%	20	56.01	65.76	2	21	+16%
2021-07	CMI	Industrials	20%	20	242.13	500.09		47	+133%
2021-07	MU	Technology	20%	20	80.11	884.74		222	+1033%
2021-07	HUM	Health care	19%	19	447.63	414.50		19	-2%
2022-01	MU	Technology	21%	21	95.75	884.74		195	+845%
2022-01	SNA	Industrials	20%	20	213.32	355.66		39	+89%
2022-01	CMI	Industrials	20%	20	219.53	500.09		50	+154%
2022-01	HUM	Health care	20%	20	465.45	414.50		18	-6%
2022-01	INTC	Technology	19%	19	53.21	65.76	2	21	+20%
2022-07	NUE	Materials	22%	22	105.64	273.87		60	+175%
2022-07	SNA	Industrials	20%	20	198.24	355.66		40	+101%
2022-07	MU	Technology	20%	20	53.65	884.74		335	+1585%
2022-07	INTC	Technology	19%	19	36.34	65.76	3	30	+73%
2022-07	CMI	Industrials	19%	19	194.11	500.09		54	+183%
2023-01	MU	Technology	22%	22	50.37	884.74		384	+1683%
2023-01	NUE	Materials	21%	21	131.06	273.87		46	+120%
2023-01	CTSH	Technology	19%	19	57.59	65.11		23	+21%
2023-01	INTC	Technology	19%	19	26.73	65.76	4	40	+130%
2023-01	SNA	Industrials	19%	19	229.38	355.66		33	+71%
2023-07	STLD	Materials	22%	23	110.78	250.80		53	+137%
2023-07	NUE	Materials	21%	22	165.61	273.87		37	+73%
2023-07	MU	Technology	21%	21	63.90	884.74		299	+1303%
2023-07	SNA	Industrials	19%	20	286.86	355.66		27	+35%
2023-07	PKG	Materials	18%	19	132.61	256.82		39	+111%
2024-01	STLD	Materials	22%	22	118.91	250.80		47	+119%
2024-01	NUE	Materials	21%	21	175.43	273.87		34	+62%
2024-01	SWKS	Technology	20%	20	109.11	87.58		18	-12%
2024-01	SNA	Industrials	19%	19	288.74	355.66		26	+32%
2024-01	CTSH	Technology	18%	18	74.40	65.11		17	-8%
2024-07	NUE	Materials	22%	25	157.00	273.87		46	+80%
2024-07	STLD	Materials	20%	24	129.15	250.80		47	+100%
2024-07	SWKS	Technology	20%	23	106.40	87.58		21	-11%
2024-07	SNA	Industrials	20%	23	256.90	355.66		34	+47%
2024-07	BG	Staples	19%	22	107.32	96.64		21	-4%
2025-01	NUE	Materials	21%	23	114.48	273.87		56	+145%

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Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2025-01	SWKS	Technology	21%	22	88.45	87.58		23	+5%
2025-01	STLD	Materials	20%	21	111.93	250.80		49	+129%
2025-01	BG	Staples	19%	20	77.57	96.64		26	+31%
2025-01	AMAT	Technology	19%	20	163.87	390.09		49	+141%
2025-07	SWKS	Technology	22%	23	77.16	87.58		27	+18%
2025-07	ON	Technology	21%	22	53.60	71.67		29	+34%
2025-07	SNA	Industrials	20%	21	317.86	355.66		24	+16%
2025-07	AMAT	Technology	19%	20	183.76	390.09		44	+114%
2025-07	STLD	Materials	19%	20	130.16	250.80		38	+95%
2026-01	SWKS	Technology	21%	23	64.40	87.58		32	+39%
2026-01	ON	Technology	21%	23	56.70	71.67		28	+26%
2026-01	SNA	Industrials	20%	22	350.95	355.66		22	+3%
2026-01	NUE	Materials	19%	21	169.40	273.87		35	+63%
2026-01	CTSH	Technology	19%	20	81.28	65.11		17	-18%
2026-07	CTSH	Technology	21%	22	41.07	65.11		34	+59%
2026-07	SWKS	Technology	20%	21	65.93	87.58		27	+33%
2026-07	SNA	Industrials	20%	20	403.28	355.66		18	-11%
2026-07	RMD	Health care	19%	20	201.10	229.94		23	+15%
2026-07	BF-B	Staples	19%	20	25.83	28.71		22	+12%

B Wallet contents by year

Table 9: Wallet with sells of the investor who started in 2020: value of each holding after each January's trades and today (USD), and the money put in and the proceeds of sales to date, July decisions included.

Holding	2020	2021	2022	2023	2024	2025	2026	Now
CMI	18	23	61	93	96	141	217	210
CTSH	21	77	88	77	119	124	154	160
MU	20	75	142	115	218	232	840	2,357
NUE	21	69	151	226	351	274	433	706
SNA	20	43	97	150	234	309	377	408
HAS		47	53	33	25	5	5	5
HUM		19	62	67	63	34	36	58
INTC			39	54	99	42	67	112
PKG					23	33	32	39
STLD					46	86	164	236
SWKS					20	59	87	148
BG						36	45	48
AMAT						20	63	92
ON							46	58
BF-B								22
RMD								23
Put in	100	300	500	700	900	1,100	1,300	1,400
Sold	0	0	0	0	4	29	43	45
Total	100	353	693	816	1,293	1,396	2,567	4,681

C Testing the weighting

Table 10: Each rule's weight w_r on the picks' dates (overall, for the picks that beat the S&P 500 and for the others), the dollar-weighted mean points, the gap and the rank correlation with the return a year above the S&P 500.

Rule	w_r	w_r beat	w_r not	Pts beat	Pts not	Gap	Corr.
NN	1.00	1.00	1.00	4.15	3.71	+0.44	-0.07
E6	0.99	0.99	1.00	4.55	3.47	+1.08	+0.51
G7	0.91	0.91	0.92	4.50	3.52	+0.99	+0.40
G6	0.88	0.87	0.88	4.62	3.57	+1.05	+0.36
G2b	0.83	0.83	0.84	4.98	4.84	+0.14	+0.10
E2	0.82	0.81	0.82	4.98	4.84	+0.14	+0.10
B1	0.79	0.75	0.85	2.32	3.73	-1.42	-0.30
G2a	0.76	0.76	0.77	4.90	4.65	+0.25	+0.22
E1	0.59	0.59	0.60	4.90	4.65	+0.25	+0.22
NI1	0.39	0.38	0.39	2.81	2.59	+0.22	-0.12
G4	0.37	0.37	0.37	0.52	1.92	-1.40	-0.19
G3	0.27	0.26	0.29	0.69	3.04	-2.36	-0.31
E5	0.25	0.25	0.25	3.55	2.75	+0.81	+0.24
E3	0.20	0.20	0.21	2.00	3.00	-1.00	-0.29
E4	0.19	0.19	0.19	1.84	2.00	-0.16	-0.07
G5	0.18	0.18	0.19	2.98	4.14	-1.15	+0.04

Each rule's points are multiplied by its weight, one less the share of companies passing it on the date, so a rule few companies pass counts more. A good weighting gives most weight to the rules whose points went with better returns. Table 10 sets each rule's mean weight on the dates the picks were made against two measures of how well it told the 110 picks apart: the gap between the mean points of the picks that beat the S&P 500 and of those that did not, weighted by dollars as in Figure 4, and the rank correlation of the rule's points with each pick's return a year above the S&P 500. Figure 9 plots weight against gap; a good weighting puts its rules on a rising line, with the heavy rules above zero.

Across the rules, the rank correlation of weight and gap is +0.61, and of weight and correlation +0.49.

The weighting leans towards the rules that worked, though far from fully. Weighted heavily but with a negative gap: return on equity (B1). A positive gap but a light weight: current ratio (lenient) (E1), 4-year earnings growth (E5) and net income growth (NI1). A rule's weight hardly differs between the dates of winning and losing picks (columns w_r beat and w_r not), because it is set across all companies on a date, not per pick. These differences suggest where the weighting could be tuned; any change must be tested on years it was not fitted on.

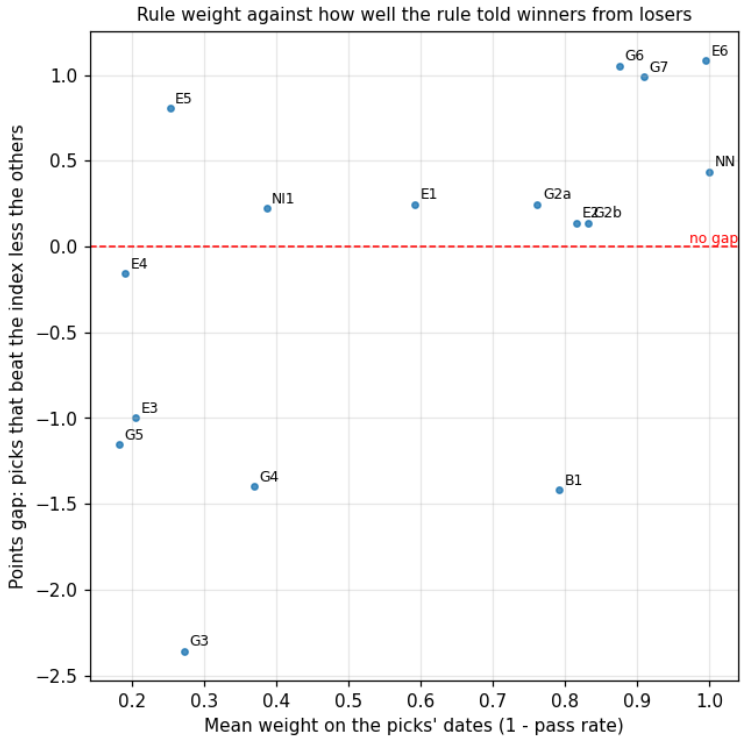


Figure 9: Each rule's mean weight against the gap in points between the picks that beat the S&P 500 and the others, weighted by dollars; rules above the line told winners from losers.

D Forward information

Table 11: Truncation check: companies scored on the date, companies whose score differs when every input is cut at the date (per score), and whether the five picks are the same.

Date	Companies	Score	Score 0 to 1	Unweighted	Share passed	Same picks
2018-12-21	296	0	0	0	0	yes
2022-12-16	339	0	0	0	0	yes
2025-12-19	376	0	0	0	0	yes

A score must use only what was known on its date. Four guards are built in. A company's figures count from the day its annual report was filed with the SEC, and of all the figures filed for a year only the first reported is used, never a later restatement. Prices are the close on the scoring date and dividends are counted by their ex-date. Each January and July purchase uses the scores of the rebalance date two weeks before it. Rule weights come from the companies on the date alone.

These guards are tested by cutting every input at a date as if it were that day (filings filed on or before it, prices and dividends up to it), scoring the companies again and comparing the scores with the full run (Table 11). Any difference means information from after the date reached a score. On 21 December 2018, 16 December 2022 and 19 December 2025 the picks were the same, and the scores were identical on 21 December 2018, 16 December 2022 and 19 December 2025. The same check runs as a unit test. A rule whose window reaches back further than a company's price history (the 20-year dividend record of a younger company) is not evaluated rather than failed.

Three sources of forward information are outside this test. Only today's S&P 500 members are scored: companies join the universe only from the date they entered the index, but companies that later left it are missing, which flatters every result. Sectors are today's, used to leave out Financials and Real Estate in earlier years. And the score's design and settings were chosen while looking at

2016 to 2026, so no period tests it fairly: 2022 onwards, once meant as a held-out test, was used to choose between models.

D.1 Companies missing from the data

Table 12: Companies that left the S&P 500 since each start year, all missing from the data: in all, on their market value, and through an acquisition or another reason.

Start	Left the index	On market value	Acquired or other
2016	230	97	133
2020	125	75	50
2023	69	47	22
2025	37	24	13

Table 12 counts the companies that left the S&P 500 after each start year; none of them is in the data, of any sector. Most left through an acquisition, often at a premium; those that dropped out on their market value are the likely losers a real investor could have picked. The later the start, the fewer are missing, which is why the investor who started in 2020 is the reference of this report. The bias is removed only when the historical members of the index are added to the data.

E The rules

Table 13: The rules: what each measures, the better direction and when a company passes.

Rule	Name	What it measures	Better	Passes when
G2a	current ratio	Current assets divided by current liabilities: what the company owns against what it owes within a year.	higher	at least 2.0
G2b	debt to working capital	Long-term debt divided by net current assets (current assets less current liabilities).	lower	at most 1.0
G3	10 years of profit	Number of the last 10 fiscal years with positive earnings per share.	higher	all 10 years
G4	20 years of dividends	Number of the 20 calendar years before the decision in which a dividend was paid.	higher	all 20 years
G5	10-year earnings growth	Average earnings per share of the last 3 years over the average of 3 years a decade earlier.	higher	at least 1.33
G6	P/E	Price over the average earnings per share of the last 3 years.	lower	at most 15.0
G7	P/E times P/B	P/E as in G6 times the price over book value per share.	lower	P/B at most 1.5, or the product at most 22.5
E1	current ratio (lenient)	Current ratio, as G2a.	higher	at least 1.5
E2	debt to working capital (lenient)	Long-term debt over net current assets, as G2b.	lower	at most 1.1
E3	5 years of profit	Number of the last 5 fiscal years with positive earnings per share.	higher	all 5 years
E4	current dividend	Dividends paid over the last 365 days, as a yield on the price.	higher	any dividend
E5	4-year earnings growth	Change in earnings per share over 4 years, relative to the earlier figure.	higher	earnings per share grew
E6	price to tangible assets	Market value over net tangible assets (equity less goodwill and other intangible assets).	lower	at most 1.2
NN	net-net	Market value over net current asset value (current assets less all liabilities).	lower	at most 0.67
B1	return on equity	Number of the last 10 years with return on equity (net income over equity) of at least 15%.	higher	at least 7 years, and long-term debt at most 0.5 times equity
NI1	net income growth	Growth of net income from the year before to the latest year.	higher	net income grew

Table 13 explains each rule: what it measures, whether a higher or a lower value earns more points, and when a company passes it. The points come from the company's rank on the measure among all companies on the date; the pass threshold sets the rule's weight. Rules G follow Graham's defensive investor (The Intelligent Investor, chapter 14), rules E his enterprising investor (chapter 15) and his net-net test (NN), B1 Buffett's return on equity, and NI1 was added for net income growth. Two pairs test the same measure at different strictness (G2a and E1, G2b and E2), so the current ratio and debt count twice.